

Practical No. 7

Aim: To create and manage users, groups, and permissions using AWS Identity and Access Management (IAM) in order to control secure access to AWS services.

Theory:

AWS Identity and Access Management (IAM) is a service that enables you to securely control access to AWS resources. Using IAM, you can create and manage users, groups, and permissions to allow or deny access to specific AWS services and resources.

IAM follows the principle of least privilege — users are granted only the permissions they need to perform their job. Permissions are defined using policies, which can be attached to users or groups. Managed Policies are reusable pre-built policies, while Inline Policies are assigned to a single user or group for one-off situations.

Key Concepts

- Users — Individual IAM identities with specific credentials and permissions
- Groups — Collections of users that share the same permissions
- Policies — JSON documents that define allowed or denied actions on AWS resources
- Roles — Temporary permissions that can be assumed by AWS services or users

Advantages of IAM

- Centralized control over AWS account access
- Fine-grained permissions at the resource level
- No additional charge — IAM is available at no cost within AWS

Steps:

Task 1: Explore Users and Groups

1. In the AWS Console, search for and open IAM. In the left navigation pane, choose Users. Three users are available: user-1, user-2, and user-3.

☐	User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key ID
☐	awsstudent	/	Access denied	-	Access denied	Access denied	-	Access denied
☐	user-1	/spl66/	0	-	-	3 minutes	-	Active - AKIA6GBMFO...
☐	user-2	/spl66/	0	-	-	3 minutes	-	Active - AKIA6GBMFO...
☐	user-3	/spl66/	0	-	-	3 minutes	-	Active - AKIA6GBMFO...

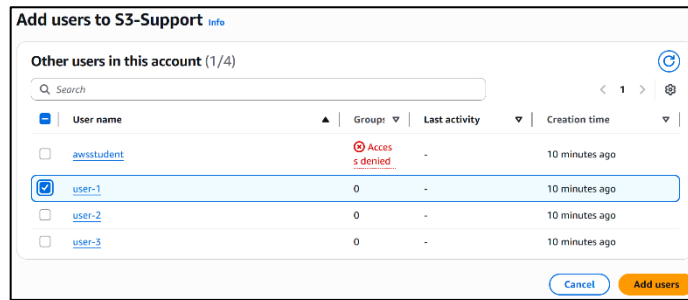
2. Click user-1 and observe that it has no permissions and is not a member of any group. It has a Console password assigned under Security credentials.
3. In the left pane, choose User groups. Three groups exist: EC2-Admin, EC2-Support, and S3-Support.

User groups (3) Info			
A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.			
Search			
☐	Group name	Users	Permissions
☐	EC2-Admin	0	Defined
☐	EC2-Support	0	Defined
☐	S3-Support	0	Defined

4. Review the Permissions tab of each group. EC2-Support has AmazonEC2ReadOnlyAccess, S3-Support has AmazonS3ReadOnlyAccess, and EC2-Admin has an Inline Policy allowing Describe, Start, and Stop actions on EC2 instances.

Task 2: Add Users to Groups

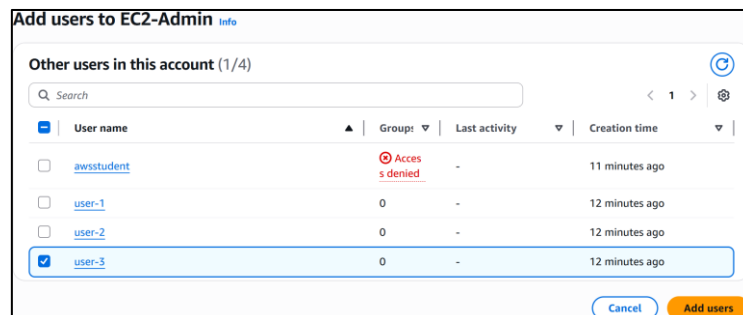
1. Open the S3-Support group, go to the Users tab, and click Add users. Select user-1 and confirm.



2. Open the EC2-Support group and add user-2 using the same steps.



3. Open the EC2-Admin group and add user-3. Verify that each group now shows 1 user in the Users column.



Task 3: Sign In and Test Users

1. From the IAM Dashboard, copy the Sign-in URL for IAM users and open it in a private browser window.
2. Sign in as user-1 (Lab-Password1). Verify that S3 buckets are visible but EC2 access is denied — confirming S3 read-only permissions.

IAM user sign in

Account ID or alias (Don't have?)
975050208992

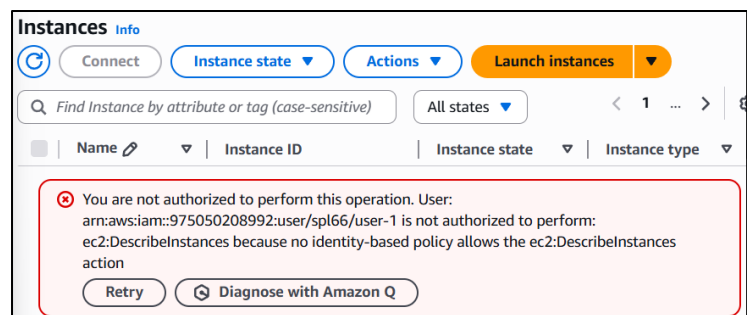
☐ Remember this account

IAM username
user-1

Password

☐ Show Password [Having trouble?](#)

Sign in



3. Sign out and sign in as user-2 (Lab-Password2). Verify that EC2 instances are visible but stopping an instance is denied — confirming EC2 read-only permissions. Also verify that S3 access is denied.

IAM user sign in

Account ID or alias (Don't have?)

975050208992

☐ Remember this account

IAM username

user-2

Password

☐ Show Password [Having trouble?](#)

Sign in

Buckets

General purpose buckets All AWS Regions Directory buckets

General purpose buckets (0) Info

Copy ARN Empty Delete Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

< 1 >

Name AWS Region Creation date

You don't have permissions to list buckets

After you or your AWS administrator has updated your permissions to allow the s3:ListAllMyBuckets action, refresh this page. [Learn more about Identity and access management in Amazon S3](#)

Diagnose with Amazon Q

Failed to stop the instance i-00d86f59581aa1571

You are not authorized to perform this operation. User: arn:aws:iam::975050208992:user/spl66/user-2 is not authorized to perform: ec2:StopInstances on resource: am:aws:ec2:us-east-1:975050208992:instance/i-00d86f59581aa1571 because no identity-based policy allows the ec2:StopInstances action.

Encoded authorization failure message: ecOG_U-wk4rNWYTIdMDmy9T_seEkhyV8QG9zsiqzo_J3DytMxJf9DaellvE_24OM2rfb4bas2STKMh5oH_EQJR82tyue1T8K0_IryX6tzqV0d7LsOsSETG3fu_fh9LOE6kPys7tkJAL-l6ivm3xHzRS9pAtWcWuGMH2e1CECaFa5YezsCuzRWht0Tx-069HozLYcRfWEDXYsAd3KMcGnNf4XSHzY3YvyBUIO1_5tNhu-qdySYa4pK3NdT28g5ct87YJBhpJHxQhN_urgMSwLGB8Pec0_sk1Hznth4H3on0KJlIXLcozu76_EQfoXC3InjQtm2g2KyyxWFEIGkp-IRbtaz5R3uUMm6dLmaMin1KVgE52ajmJ8oDeuZjC-F5WSOTYbt9A4Lq-MJPJ16pyy1QaRVCBNLFH8Rqbg3B9nmCW0TzgPQE5v3hsPQSUF5JERH-xYh2mCemuXvZ7B58Vtzw_t4UxRgAVm4Bb1IV4_VNJhTHaAMECRou7UloBgO7kQpxlvf_-RMVwkvuW_aCL3n4VrvkMYI9w-kKQTLbR5Wou5lIF6PD3UBUgdjt8Dj9ndvCIBy59-RAIES0bJBNPCcpBmH5S5XnqrUoEubW71zBk6FcxWZk64YILF2cPYD8gqGN8pMisPX5XWTDKya51Xa7CUYQu-lpw7n60mFIX5pUNoonPi8KI4GncgmRZO8EB47s-csDhqIV11w8jklqRiBPwe6aD26BKpnmvqC4UpPxRQTU7TIX5lyJS2Rf0Vkm5WKH4wxt3mpR502Sc7gYpekQyv5gZ7lGuqSf5syscoGODOHHUfu7xGJWBxyFXPIebWzjheR5hMhUJcm3DxxFBZ-13_bc--PbXAGH4bUTt8vga98gnqMykv636I89ObdLLmBKJjuM7EdhovjFujzRtQ9bw8L22FNl9QaGDFvTTbF2Q90TTLk4X57cv6N_8KGxackrIPKZJHyLEgo2wiZIGSvhVf03EeQVZIDHg6ZrPg2WzRJQKe8FDB-3uG1YxWfGpm_ateaC-AZxJHFaFE49TtrM2DJud8RE3fg5NG_1n8WNvcU10

Diagnose with Amazon Q

Instances (1/2) Info

Find Instance by attribute or tag (case-sensitive)

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☐	Bastion Host	i-0218dab8c93cdc50d	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-54-196-119-51.co...
☒	LabHost	i-00d86f59581aa1571	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-3-80-192-49.comp...

4. Sign out and sign in as user-3 (Lab-Password3). Verify that stopping the EC2 instance named LabHost is successful — confirming EC2 Admin permissions.

IAM user sign in

Account ID or alias (Don't have?)

975050208992

☐ Remember this account

IAM username

user-3

Password

Lab-Password3

☒ Show Password [Having trouble?](#)

Sign in

Stop instance

Stopping your instance allows you to reduce costs, modify settings, and troubleshoot problems.

Instance ID Stop protection Result

i-00d86f59581aa1571 (LabHost) Disabled Can stop

Associated resources

You will continue to incur charges for these resources while the instance is stopped

You will be billed for associated resources

After you stop the instance, you are no longer charged usage or data transfer fees for it. However, you will still be billed for associated Elastic IP addresses and EBS volumes.

Skip OS shutdown

This option skips the graceful OS shutdown process. Use only when your instance must be stopped immediately, such as during an emergency or failover.

☐ Skip OS shutdown

Cancel **Stop**

Conclusion: Thus, we have successfully created and managed users, groups, and permissions using AWS Identity and Access Management (IAM) to control secure access to AWS services.